

Apollo

Short audio clip $\longrightarrow$ song title + source time

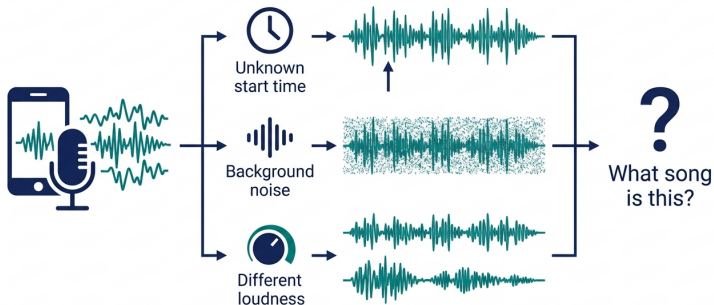
Noisy recordings, speed edits and covers

Presenter A & Presenter B

CSE 220 final project | September 2026

Why can't we just compare waveforms?

audio fingerprinting problem



We need features that survive these distortions.

Matching raw samples fails: the start is unknown, noise changes amplitudes, and every recording

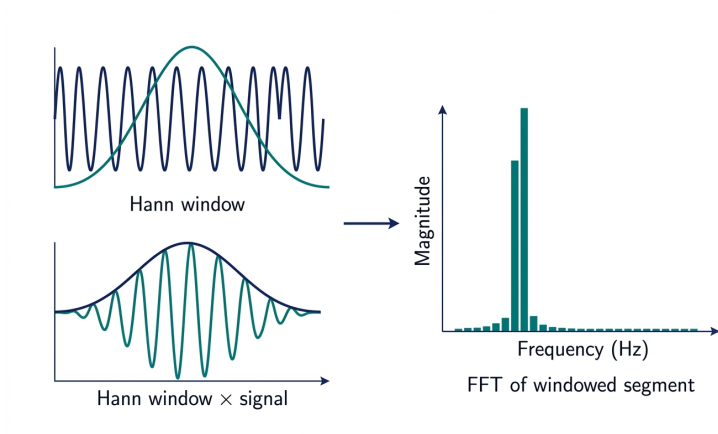
How a clip becomes an answer



Answer back to the page: song title + where the clip begins

Signal code builds fingerprints; SQLite searches an indexed catalog.

Windowing: one window, one frequency snapshot

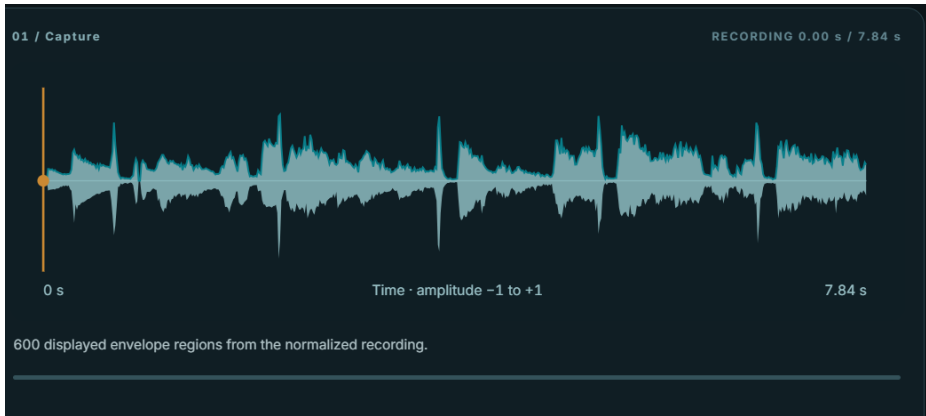


A **Hann window** tapers the edges so the FFT does not see an abrupt cut. Multiplying the signal by the window and computing the FFT gives one column of the spectrogram: *which frequencies are strong right now*

From sound to spectrogram

1 / 2 Capture the waveform

time →; height = signal amplitude

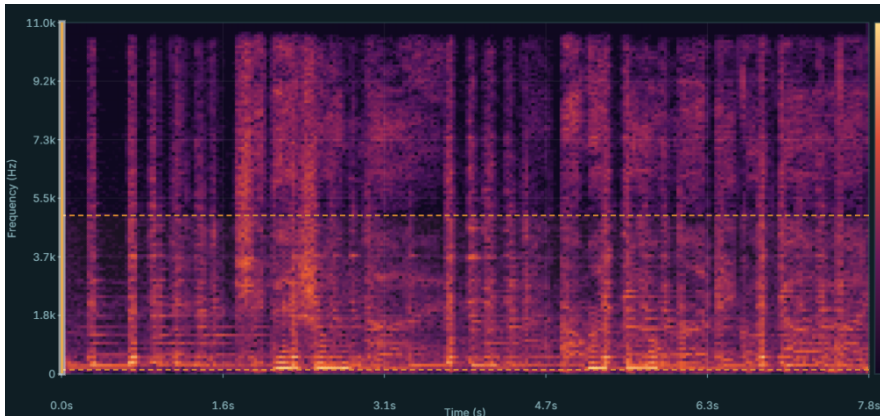


Mono + peak normalization + 22,050 samples/second for both song and query.

From sound to spectrogram

2 / 2 STFT makes a frequency map

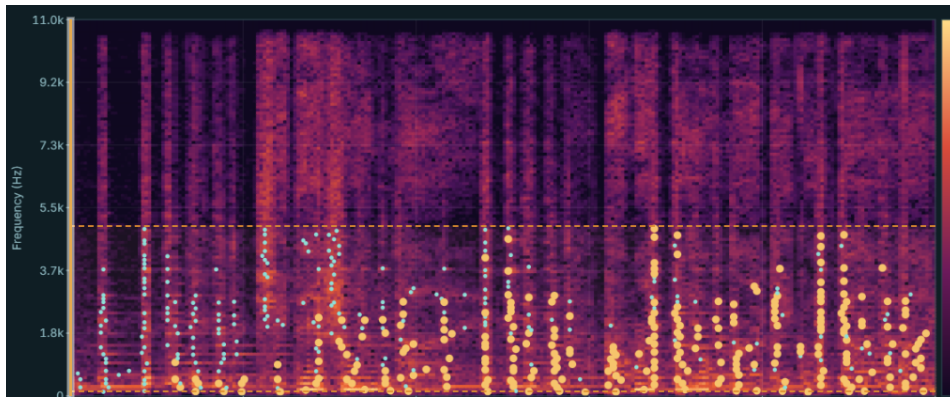
time $\rightarrow$, frequency $\uparrow$



2,048-sample windows (93 ms); hop 512 (23 ms). Each window gives one column.

Keep strong local peaks

Each dot marks one frequency at one moment.



15 frequency bins $\times$ 9 time frames · 100–5,000 Hz; ≥ -60 dB · at most 60 peaks/second

How Apollo handles edited audio

	Frequency / pitch	Time between peaks
Speed edit	changes	changes
Pitch-only edit	changes	stays the same

78 quick guesses: 39 speed + 39 pitch-only factors

Rescale detected peaks → make new fingerprints → test the best guesses.

This fallback runs when ordinary fingerprint matching misses an edit.